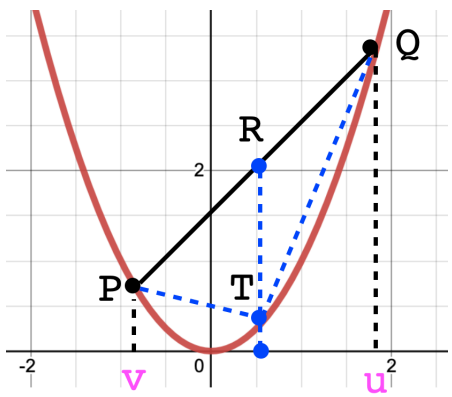
# Archimedes' Quadrature

Quadrature refers to area, in particular to a method that can find the area of a plane figure. Consider the parabola $y = x^2$ with its vertex at the origin. (For simplicity we will not worry yet about the shape factor $a$).




Pick any two points on the parabola $P$ and $Q$, then find the vertical line $RT$ that lies exactly halfway in between.

Let the $x$-coordinates of $Q$ and $P$ be $u$ and $v$, respectively. Then, the $y$-coordinates of $Q$ and $P$ are $u^2$ and $v^2$.

Notice that $RT$ produces two smaller triangles of equal area, since they have the common base $RT$ and equal height. That height is half the horizontal distance between $P$ and $Q$

$$h = \frac{u - v}{2}$$

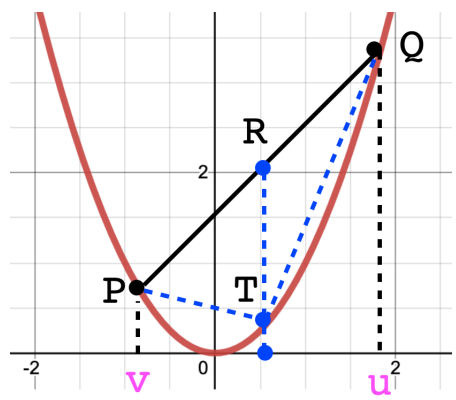
It is a moderate challenge to find the length of $RT$. The $x$-coordinates of the points $R, T$ and the intersection of $RT$ with the $x$-axis are all the same: the average of $u$ and $v$ or $(u+v)/2$.

Since $T$ is also on the curve its $y$-coordinate is the square of the $x$-coordinate $[(u+v)/2]^2$.

The $y$-coordinate of $R$ is the average of those for $P$ and $Q$ or $(u^2+v^2)/2$.

Finally, the length of $RT$ is difference of those two $y$-coordinates:

$$b = \frac{u^2+v^2}{2} - [\frac{u+v}{2}]^2$$




Before proceeding to the area calculation, we can simplify this expression.

$$\begin{aligned} 4b &= 2u^2 + 2v^2 - u^2 - 2uv - v^2 \\ &= u^2 - 2uv + v^2 = (u-v)^2 \end{aligned}$$

The area of each small triangle is $hb/2$ so the area of $\triangle PQT$ is

$$\begin{aligned} A = hb &= \frac{u-v}{2} \cdot \frac{(u-v)^2}{4} \\ &= \frac{1}{8}(u-v)^3 \end{aligned}$$

That's a pretty simple result.

We ask, what would happen if we repeat the process? Subdivide $PT$ and $TQ$. Let $w$ be the $x$-coordinate of the original bisector. Then the new distance is $u - w = (u - v)/2$.

$$A' \stackrel{?}{=} \frac{1}{8}(\frac{u-v}{2})^3$$

This is not quite right yet. The reason is that we have here the area for the two new triangles between $T$ and $Q$. But there are also two new triangles between $P$ and $T$. So actually

$$A' = 2 \cdot \frac{1}{8}(\frac{u-v}{2})^3$$

$$A' = 2 \cdot \frac{1}{8} \cdot \frac{1}{8}(u-v)^3$$

$$A' = \frac{1}{4}A$$

Doing it yet again would yield

$$A'' = \frac{1}{4}A' = \frac{1}{16}A$$

Thanks to a suggestion from Eudoxus of Cnidus, we repeat this until the area just above the curve is *exhausted*:

$$\Delta A = A' + A'' + \cdots =$$
$$= \ [\ \frac{1}{4} + \frac{1}{16} + \ldots\ ]\ A$$

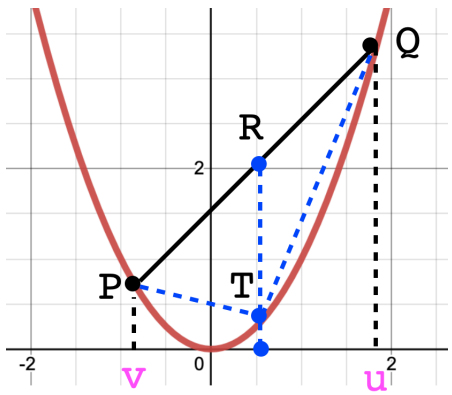
This is a geometric series with $r = 1/4$ and first term $a = 1/4$ and its sum is known to be

$$\frac{a}{1-r} = \frac{1/4}{3/4} = \frac{1}{3}$$

so

$$\frac{A + \Delta A}{A} = \frac{4}{3}$$

The total area between the curve and the secant $PQ$ is equal to $4/3$ times the area of $\triangle PTQ$.




**checking our work**

We check this with a combination of simple calculus and not-so-simple algebra. The area below the curve is the integral:

$$\int_v^u x^2 \, dx = \frac{1}{3}(u^3 - v^3)$$

We find a factor of $u - v$ in the above equation:

$$A_\int = \frac{1}{3}(u - v)(u^2 + uv + v^2)$$

Earlier, we determined that the $\triangle PTQ$ has area

$$\frac{1}{8}(u - v)^3$$

The trapezoid or box formed by $PQ$, the verticals, and the $x$-axis has

for its average height the $y$-value of the point $R$:

$$h = \frac{u^2 + v^2}{2}$$

and width $u - v$ so its area is

$$A_{\text{box}} = (u - v) \cdot \frac{u^2 + v^2}{2}$$

The areas are

$$A_{\text{box}} = A_{\int} + \frac{4}{3} A_{\triangle}$$

We factor $u - v$ from each before re-writing the last equation as

$$\frac{u^2 + v^2}{2} = \frac{1}{3}(u^2 + uv + v^2) + \frac{4}{3} \cdot \frac{1}{8} \cdot (u - v)^2$$

Working with the right-hand side, we clearly we have a factor of 1/3, and that has me a little worried as there certainly is not one on the left.

$$= \frac{1}{3}(u^2 + uv + v^2 + \frac{(u - v)^2}{2})$$

We have from that second term $(u^2 - 2uv + v^2)/2$ which gives not only a cancelation but also factors of 3/2:

$$= \frac{1}{3} \cdot \frac{3}{2}(u^2 + v^2) = \frac{(u + v)^2}{2}$$

And that's a relief. ✓

**halfway point**

The slope of the tangent to the curve at point $T$ is twice the $x$-coordinate of $T$, times $a$ if present (see below). That's just $2 \cdot (u+v)/2 = u + v$.

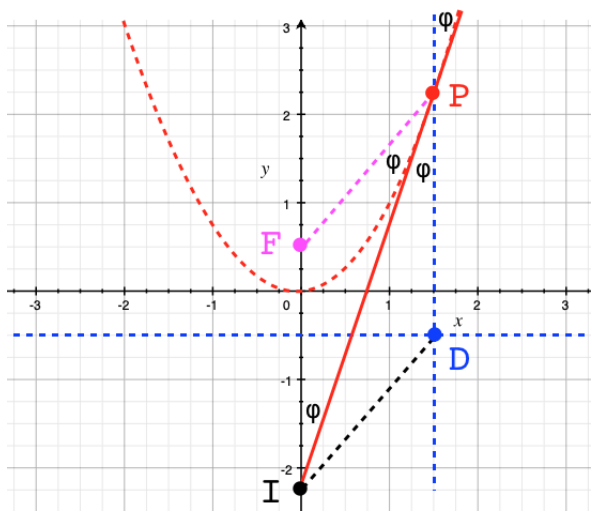
The slope of the secant $PQ$ from analytic geometry is the same:

$$\frac{\Delta y}{\Delta x} = \frac{u^2 - v^2}{u - v} = u + v$$

**slope of the tangent**

At this point, you might ask just why it is that if we pick two points and find the third point whose $x$ is midway, that the slope of the secant through the two points is equal to the slope of the curve at $x$.

A French mathematician named Roberval had a beautiful answer for that.




The geometric definition of the parabola is to draw a line called the directrix and also pick a point $F$, the focus. Then every point on the parabola $P$ has the same distance to $F$ as vertically down to $D$ on the directrix.

Since the direction of "motion" must keep these distances the same, it follows that the tangent bisects $\angle FPD$. It follows that $FPDI$ is a parallelogram.

As a result the slope of the tangent line $PI$ is $2ax^2/x = 2ax$.

A second approach comes from calculus. Literally the first thing we do is to find the slope of the secant for two nearby points (separated by $\Delta x$), that satisfy a quadratic equation $y = ax^2$.

$$\begin{aligned} f'(x) &= \frac{a(x+\Delta x)^2 - ax^2}{\Delta x} \\ &= \frac{2ax\Delta x + (\Delta x)^2}{\Delta x} \\ &= 2ax + \Delta x \end{aligned}$$

The magic is that we can let $\Delta x$ "tend" to zero (despite having it in the denominator the moment before), and then this becomes the slope of the tangent line, which is $2ax$.

A third approach is to write the equations for a line that intersects a parabola.

$$\begin{aligned} y &= kx + y_0 \\ y &= ax^2 \end{aligned}$$

So

$$ax^2 - kx - y_0 = 0$$

The roots are

$$x = \frac{k \ \pm \sqrt{k^2 - 4ay_0}}{2a}$$

It is possible that there are no solutions, one solution, or two. The case of one solution is the tangent line, and that happens when the discriminant is zero.

$$\begin{aligned} x &= \frac{k}{2a} \\ k &= 2ax \end{aligned}$$

**parallel lines**

In *Infinite Powers*, Strogatz describes this problem from the reverse direction. He says that Archimedes starts with the secant and then moves that line parallel to find where it just touches the curve.

We have our two points and a line with slope

$$\frac{u^2 - v^2}{u - v} = u + v$$

and now we want the intersection(s) of a line with this slope and the curve

$$y = x^2 = (u + v)x + y_0$$

We want the point where there is a single solution.

Although it seems that we must find $y_0$, we don't have to, because it is a constant. For the sake of completeness, though, we will. Plug the point $(u, u^2)$ into the equation of the line

$$\begin{aligned} y &= (u + v)x + y_0 \\ u^2 &= (u + v)u + y_0 \\ y_0 &= -uv \\ y &= (u + v)x - uv \end{aligned}$$

You can easily confirm that both $(u, u^2)$ and $(v, v^2)$ satisfy this equation.

Solving our problem

$$\begin{aligned} y = x^2 &= (u + v)x - uv \\ x^2 - (u + v)x + uv &= 0 \end{aligned}$$

This is a quadratic in $x$ which has a single solution when

$$x = \frac{u + v}{2}$$

$x$ lies halfway between $u$ and $v$, as we said.

**factor of $a$**

A factor of $a$ in the equation $y = ax^2$ stretches all $y$-values and all the areas in the $y$-direction. The simplest way to deal with this is to do a variable substitution $y/a = u$ and then follow what we did here.

In interpreting the areas, every area will have a factor of $a$ to add back, from the stretching, and those will cancel. We leave that as an exercise.